
Memory-Driven Mixed Low Precision Quantization For Enabling Deep Network Inference On Microcontrollers

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Abstract

This paper presents a novel end-to-end methodology for enabling the deployment of low-error deep networks on microcontrollers. To fit the memory and computational limitations of resource-constrained edge-devices, we exploit mixed low-bitwidth compression, featuring 8, 4 or 2-bit uniform quantization, and we model the inference graph with integer-only operations. Our approach aims at determining the minimum bit precision of every activation and weight tensor given the memory constraints of a device. This is achieved through a rule-based iterative procedure, which cuts the number of bits of the most memory-demanding layers, aiming at meeting the memory constraints. After a quantization-aware retraining step, the fake-quantized graph is converted into an inference integer-only model by inserting the Integer Channel-Normalization (ICN) layers, which introduce a negligible loss as demonstrated on INT4 MobilenetV1 models. We report the latency-accuracy evaluation of mixed-precision MobilenetV1 family networks on a STM32H7 microcontroller. Our experimental results demonstrate an end-to-end deployment of an integer-only Mobilenet network with Top1 accuracy of 68% on a device with only 2MB of FLASH memory and 512kB of RAM, improving by 8% the Top1 accuracy with respect to previously published 8 bit implementations for microcontrollers.

1 Introduction

Enabling machine learning on extreme-edge-devices is challenging due to their tight memory and computing power constraints. When envisioning smart sensors operating on batteries, the target power envelope must be below tens of mWs to guarantee a battery lifetime of years. This requirement impacts the system architecture design: adding computational units (e.g. floating-point units) or memory banks contributes increasing the complexity and the power cost, and hence the energy, of a system.

Nowadays, microcontroller units (MCUs), such as STMicroelectronics STM32 devices, feature an energy consumption compliant with the requirement of smart autonomous sensors and include energy-efficient computational units for running machine learning workloads. However, the typical size of the embedded memory cuts is limited to a few MB (a STM32H7 MCU features 2MB of FLASH memory) and the computation core (commonly a single ARM Cortex-M CPU) runs up to few hundreds of MHz. To boost the performance of this class of MCUs while leveraging the high

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